

Supplementary Material

Prefix-Causal Harm-Budget Projection for Cross-Domain Lithium-Ion Battery Capacity-Retention Estimation

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1 Battery datasets and engineering heterogeneity

Tables S1 and S2 report the engineering attributes encoded in the frozen processed tables rather than inferring missing values from model behavior. Temperature is the range available in the scored artifact (measured early-charge temperature or ambient set point, as indicated by the source); NR means that the field was not available in that artifact. The development metadata originate from the BatteryLife processing contract and its archived snapshot [7, 8]. The external rows were checked against the six cited primary dataset records [9, 1, 4, 3, 2, 6]. Rates and state-of-charge windows describe the protocol metadata represented in the scored records; they are not universal specifications for every experiment in the source archive.

Table S1. Engineering characteristics of the twelve development domains. Counts refer to the post-reference scoring roster.

Domain	Cells	Records	Chemistry; form factor	Q_n (Ah)	Temp. (°C)	Charge/discharge rate; cycling SOC	Principal held-out shift represented
CALB	27	2,629	NMC/graphite; prismatic	58	0–45	1–5C / 1–15C; 0–100%	Large-format cells; rate and temperature variation
CALCE	13	14,233	LCO/graphite; prismatic	1.10–1.35	NR	0.5C / 0.5–1C; 0–100%	LCO chemistry and low-capacity prismatic format
HNEI	14	15,085	LCO–NMC blend/graphite; 18650	2.8	NR	2C / 1C; 0–100%	Blended cathode chemistry and batch variability
HUST	77	145,735	LFP/graphite; 18650	1.1	NR	5C / 2–5C; 0–100%	High-rate LFP cycling
MATR	169	139,148	LFP/graphite; 18650	1.1	24.9–42.2	1–8C / 4C; 0–100%	Diverse fast-charging policies
MICH	40	19,681	NMC111/graphite; pouch	2.36	NR	1C / 1C; 0–100%	Pouch format and NMC111 chemistry
MICH_EXP	18	6,455	NMC111/graphite; pouch	5.0	NR	0.2–2C / 0.2–2C; 0–100 or 50–100%	Capacity scale, rate, and partial-SOC protocols
RWTH	48	105,140	NMC/graphite; 18650	1.85	NR	2C / 2C; 20–80%	Partial-SOC apparent-capacity target
SDU	86	48,716	NMC532/graphite; cylindrical	2.4	24.4–32.2	Rate NR; 0–100%	NMC532 source with unavailable rate metadata
SNL	61	96,092	LFP/NCA/NMC; 18650	1.1–3.2	14.3–46.3 ^a	0.5C / 0.5–3C; 0–100 or 20–80%	Multi-chemistry, temperature, rate, and SOC mixture
UL_PUR	10	2,195	NCA/graphite; 18650	3.4	19.7–33.2	2C / 1C; 0–100%	NCA chemistry and high charge rate
XJTU	23	6,823	NMC523/graphite; 18650	2.0	NR	2–3C / 1C; 0–100%	NMC523 chemistry and mixed charge rate

^aTemperature is available only for the scored SNL records carrying that channel. Chemistry abbreviations follow source metadata: LCO, lithium cobalt oxide; LFP, lithium iron phosphate; NCA, nickel cobalt aluminum oxide; NMC, nickel manganese cobalt oxide.

Table S2. Engineering characteristics of the six protocol-frozen external datasets. Counts refer to the evaluated post-reference records.

Dataset	Cells	Records	Chemistry; form factor	Q_n (Ah)	Temp. (°C)	Characterization/aging metadata	Principal shift from development
AMPERE A123 LFP	20	743	LFP/graphite; 18650	1.1	NR	$\frac{1}{3}C$ charge/discharge; 0–100% SOC	Independent LFP archive and characterization workflow
Imperial M50T	40	269	NMC811/graphite– SiO_x ; 21700	5.0	10–40 ambient	0.3C / 0.1C RPT; multiple SOC windows	Large 21700 cells, SiO_x anode, temperature and SOC shift
Luh NMC– SiO_x	180	2,377	NMC/graphite– SiO_x ; 18650	3.0	0–40 ambient	0–1.67C / 0–1C; multiple SOC windows	Broad calendar/cycling conditions and SiO_x anode
ILCC varying usage	244	4,211	NMC/graphite; polymer pouch	0.25	30 ambient	0.5–3C / 0.5–2.475C; varied depth of discharge	Small pouch cells and high protocol diversity
Stanford calendar aging	30	765	LCO/LFP/NCA; 18650 and pouch	0.24–3.50 ^b	24–60 ambient	0.2C RPT; calendar storage at multiple SOC	Mixed chemistry, form factor, and calendar-aging conditions
Multistage 50E	145	1,347	NCA/graphite; 21700	4.9	23–45 ambient	0.2–1C / 0.2–2C; varied partial-SOC windows	Multi-stage designed aging on large-format cells

^bNominal capacity was reconstructed by the frozen Stanford ingestion contract from the reported 0.2C characterization current; the range is therefore marked as derived rather than a single manufacturer specification. Zero-rate entries in the Luh metadata denote non-cycling/calendar conditions.

2 Evidence and sample-flow audit

Tables S3 and S4 reconstruct cohort flow directly from the frozen ingestion reports and final prediction ledgers. “Candidate records” denotes records presented to the corresponding frozen structural parser, not a universal count of every row or measurement in the source repository. NR means that the upstream report did not retain a comparable raw-record total. Such values were not inferred from downstream counts. Feature-reference eligibility requires sufficient valid charge-side history to form the first-five commissioning reference; target-reference eligibility additionally requires a valid first-five discharge-capacity reference. No cell or record was removed because of prediction error or the direction of a method comparison.

Table S3. Development-domain record flow. Every source cell remained in the final roster.

Domain	Source cells	Candidate records	Structural rejects	Feature records	First-five records withheld	Scored records
CALB	27	2,764	0	2,764	135	2,629
CALCE	13	14,298	0	14,298	65	14,233
HNEI	14	15,164	9	15,155	70	15,085
HUST	77	146,120	0	146,120	385	145,735
MATR	169	140,001	8	139,993	845	139,148
MICH	40	19,895	14	19,881	200	19,681
MICH_EXP	18	6,545	0	6,545	90	6,455
RWTH	48	105,397	17	105,380	240	105,140
SDU	86	49,146	0	49,146	430	48,716
SNL	61	96,492	95	96,397	305	96,092
UL_PUR	10	2,245	0	2,245	50	2,195
XJTU	23	6,938	0	6,938	115	6,823
Total	586	605,005	143	604,862	2,930	601,932

Table S4. External and boundary-program sample flow. Post-reference feature cells have sufficient charge-side history; target-eligible cells additionally have a valid outcome reference.

Dataset/program	Source cells	Candidate records	Feature cells	Feature records	Post-reference cells	Post-reference records	Target-eligible cells	Scored records
AMPERE A123 LFP	20	847	20	843	20	743	20	743
Imperial M50T	40	511	40	469	40	269	40	269
Luh NMC-SiO _x	228	3,463	228	3,457	180	2,377	180	2,377
ILCC varying usage	251	NR	251	NR	244	4,211	244	4,211
Stanford calendar aging	259	NR	259	NR	30	765	30	765
Multistage 50E	147	2,085	147	2,082	145	1,347	145	1,347
BaSyTec	48	NR	47	3,336	47	3,101	45	2,969
NASA	34	7,885	34	2,768	34	2,598	33	2,556

For AMPERE, four characterization members had fewer than three required Step-13 rows; Imperial excluded 42 summary rows without a matched charge member. Luh excluded six invalid checkups and 48 cells with fewer than six valid paired checkups. ILCC excluded seven cells with fewer than six valid capacities. Stanford retained 30 of 259 source cells after requiring six valid diagnostics; the ingestion report records 9,060 diagnostic exclusions due to charge-window gaps, incomplete window coverage, or fewer than seven cycles, but not a comparable raw diagnostic denominator. Multistage excluded three RPT members missing capacity, charge, or discharge fields and two cells with fewer than six valid RPTs. In BaSyTec, F0001 was reserved before outcome access for parser development; F0009 and F0048 fell outside the frozen first-five reference-capacity support interval. For NASA, 5,117 cycle records lacked a valid fixed charge window and 170 commissioning records were withheld. The independent label audit rejected 44 invalid discharge-capacity records; one predicted battery consequently lacked a valid first-five target reference, leaving its 42 prediction records unscored.

3 Feature-by-weighting attribution audit

The post-freeze 2×2 audit crossed raw versus commissioning-reference-change features with pooled versus domain-equal source weighting. The four candidates used identical strict leave-one-domain-out folds, ExtraTrees settings, source-row caps, imputers, and random seed. Each candidate was then passed through the current PCHP operator using the raw/pooled predictor as the protected state and the frozen V326 outer-domain α schedule. The selected change/domain-equal arm reproduced the V327 prediction ledger with maximum absolute error 0.

Table S5. Four-arm feature-by-weighting audit. MAE is averaged equally across cells and then across the 12 complete held-out domains.

Features	Source weighting	Candidate MAE	PCHP MAE	Worst-domain PCHP MAE
Raw	Pooled	0.07790	0.07625	0.14400
Raw	Domain-equal	0.08169	0.07588	0.14351
Reference-change	Pooled	0.05685	0.07059	0.14531
Reference-change	Domain-equal	0.05694	0.07038	0.14531

Before projection, the change-feature effect was -0.02105 under pooled weighting and -0.02474 under domain-equal weighting. Table S6 reports paired effects after PCHP, where negative values favor the named intervention. Intervals use 100,000 bootstrap resamples of complete domains; p values use all $2^{12} = 4,096$ paired sign assignments. The feature effect was consistent under both weighting schemes, whereas neither weighting main effect nor the feature-by-weighting interaction was resolved from zero.

Table S6. Paired PCHP-layer effects in the 2×2 audit.

Contrast	Mean MAE difference	95% interval	Exact p	W/T/L
Change vs raw, pooled weighting	-0.00566	$[-0.00941, -0.00235]$	0.0078	11/0/1
Change vs raw, domain-equal weighting	-0.00550	$[-0.00970, -0.00184]$	0.0103	10/0/2
Domain-equal vs pooled, raw features	-0.00037	$[-0.00166, 0.00064]$	0.6357	7/0/5
Domain-equal vs pooled, change features	-0.00021	$[-0.00058, 0.00015]$	0.3027	7/1/4
Feature-by-weighting interaction	0.00016	$[-0.00093, 0.00133]$	—	—

Table S7. SHA-256 identifiers for the V383 2×2 attribution audit.

Artifact	SHA-256
Audit program	8f674bd9b452c71af7d6080f5058d0d94047e2c90f00ad34799630a21e831518
Result report	2fa763c923b552bb87d218269f4a2266ce7ec309972e43db7fda1f14e041ab60
Candidate prediction ledger	96073dfe964f3cd119bfa5cff5622f6aec87481a84df7f3da1ec59beb5a2957a
Projected prediction ledger	09e172a62f840d4afb0d7ed45188c892e01c21503e39e5af463090d55c892098

4 Frozen artifact chain

Table S8 records selected SHA-256 identifiers that bind the prefix-causal implementation, nested development evaluation, sample-flow audit, unprotected comparator, protocol-locked BaSyTec external validation, and NASA workflow to the reported results. Original datasets remain governed by their respective repositories and licenses.

Table S8. Selected SHA-256 identifiers for the causal method, evidence-flow audit, and protocol-frozen external workflows.

Artifact	SHA-256
Prefix-causal projection implementation	ce7288a129c17114e1ca57432c6417beba7938d58db2b1fd0a87171c479eb54c
Nested development predictions	d7a9a8c06923841260d98526dff665562ca27624357a4d514f6bc75563c69534
Candidate-control implementation	0f6523700082cff38aa26cd05385d3cf3252d436ccdbcc22821355e7029861873
Source-tuned unprotected comparator report	ed11518ea3e1eaa42148bf4421f0570d775074ae697260a8edf9fbac0798c252
Sample-flow protocol	94975fc1371f28591bf69e6c82d5e3d6d2bb1677e4bfa6af4e5d97ddac25f981
Sample-flow audit program	2131b21b91991b7a73d260b7799264aea43ad296e1ddb21c8143a74d22fff3c1
Surface-level sample-flow ledger	6417dca163da4e9fe872b480f9072a6985def149008f5aea57c23fe306ce47c7
Sample-flow audit report	48c2f0f62c1a08b1939bf3808bb4a4a3e99489d3510042c94aadbb7373cdf7cfe
BaSyTec final pre-freeze protocol	2e7c34cc3e56cbd5901d18b3989d18754d08e66e7448205197e5b6aa9658643a
BaSyTec header-only inventory	35c4fa46ad72f00f09a17282e0ed4a2c04ec03c33c852ea621fafc746ab6a656
Frozen BaSyTec predictions	2e27780968f909b0d1d1a05b2b90370e2735e99f3a8e8926b781150751b6f8f6
BaSyTec single-run result report	eb6bf718810577fe6e91228301e7352236a991d9899b3002fc2ee2d5753ec2b0
BaSyTec independent statistical audit	7726061409e3a897fa2677c519a7a1af8e3ca1771fac384d8f0fed2c0682fc7
NASA pre-freeze protocol	37507d16ed8f13bd4aa378f3da3480dad691addf383c31b00ad613bb7397adea
Official NASA archive	82302a7db4fc1b34e0b6676326610438d43b816bdf11a69d1d012a464ef2f92e
Frozen NASA predictions	4d8264d94e224be12a5574cc403047e10276c78652d80a4841086f4e409ceaa9
Independent NASA scoring program	9894c1cbecae80e8a0712b812257ab474974ee3a3ea8df418e6306f94bdf40ba
NASA single-run result report	d92be40ab5c34b6324667af549cfde34b15857d02e8c0c654579111854de48b1

5 Guarantee and workflow verification

Thirty historical V321 unit tests cover causal-state monotonicity, unit assimilation, prefix invariance, nearest-current-feasible projection, range and harm-budget-interval membership, the exact harm identity and harm-region equivalence, zero-budget impossibility, recursive feasibility and exact viability-kernel membership, asymmetric-loss constraints, time-varying feasibility and minimum viable budgets, invalid settings, non-finite and shape-mismatched inputs, and cellwise order restoration. Four separate V377 unit tests cover the exact local proximal representation, protected-state and complete-operator prefix nonexpansiveness, and tightness of the unit Lipschitz constant, giving 34 focused PCHP tests without modifying the historical V321 file. Four synthetic NASA workflow

tests, five BaSyTec header-family tests, and one synthetic MATLAB integration test verify field isolation and two-stage execution. The development, BaSyTec, and NASA reports additionally check maximum displacement, observed absolute-loss increase, trajectory monotonicity, independent-unit counts, frozen file identities, and aggregate reconstruction.

6 Retrospective budget-path audit

The budget-path analysis was protocol-locked before its computation but after all development-domain outcomes had been opened. Its single primary empirical estimand is the domain-equal mean of the budget-normalized integrated PCHP-minus-exact-budget-feasible-offset MAE difference. Per-budget intervals are descriptive. The original V361 validator recorded **NARROW** because one historical anchor compared the high-precision V327 value with a report JSON rounded to ten decimal places under a 10^{-12} tolerance. V361 was not modified or rerun. A separately frozen V362 audit verified the authoritative aggregate CSV, all twelve domain rows, the V326 source-only selection roster, and all non-anchor scientific checks. It classified the anchor failure as a serialization-precision false negative while retaining the retrospective label and the original V361 decision record.

Table S9. SHA-256 identifiers for the budget-path analysis and independent anchor audit.

Artifact	SHA-256
V361 analysis pre-freeze	7bc32a42d2c20e61471357126f6281e3a11ac976979d56056b68b243f6a6c4d6
V361 analysis program	bdd0a1aeaf715d25fe574feb52b1679551a608999ab49848a3ff8c6dfd701f79
V361 result report	6e8d09554f73f1df7cdf6cf8affdaaf1c13eec80b6778b5e58616038ec02781b
V361 domain AUC table	f3c1275be42e9de2d38a4d516c3518a3777bad5a0341f4d07a2852d28693c416
V362 audit pre-freeze	49b986a77346f28c80c3d024bae6caff58de469124c422db726945c72400ea56
V362 audit program	ca3e2be80c1e995daaa90741ab6d6e07af94424a294d6b80e09f050b32640b5
V362 audit report	8e02eca68f7016d0ec08100195689e0129cccc90fc8add7b6ecc4a275b323608

7 Finite-sample domain sensitivity audit

Because the development estimands have only 12 independent complete-domain effects, V378 recomputed three paired comparisons using complementary small-sample procedures. For each comparison it enumerated all $2^{12} = 4,096$ sign assignments for the domain-equal mean, computed a Student- t interval, and reported exact paired Wilcoxon and direction-only sign tests. This audit was performed after the development outcomes had been opened. It is a sensitivity analysis, not a new confirmatory test, and it leaves the frozen estimands and percentile-bootstrap reports unchanged.

Table S10. Post-hoc finite-sample sensitivity over complete domains. Negative means favor PCHP. All test values are two-sided and descriptive.

Comparison	Mean	Student- t 95% interval	Sign-flip value	Sign-test value	W/T/L
PCHP vs protected baseline	-0.004862	[-0.007033, -0.002692]	0.000977	0.006348	11/0/1
PCHP vs matched constant-offset control	-0.004773	[-0.008581, -0.000965]	0.016602	0.145996	9/0/3
Integrated budget-path AUC	-0.006942	[-0.011718, -0.002166]	0.007812	0.006348	11/0/1

The magnitude-sensitive calculations preserve all three negative domain-equal mean directions. The direction-only sign test for the matched constant-offset comparison does not fall below 0.05, so that result supports an average utility contribution with heterogeneous domain effects, not universal dominance or a claim that improvement is more likely than not in a randomly drawn new domain.

Table S11. SHA-256 identifiers for the V378 finite-sample audit.

Artifact	SHA-256
V378 audit program	50426e90264f15e1632d0fa8403be7b16c7ef09c9076a34be05897f1a0b0b578
V378 audit report	411007452ed8c095258f3949af959e2518dc1c63e648583e5ffe9dde3b265ae9

8 Untuned secondary-backbone portability audit

This protocol-locked retrospective audit replaced both primary ExtraTrees regressors with Ridge, HistGradientBoosting, or LightGBM while preserving the data, feature contract, weighting, outer-domain exclusions, harm budget, PCHP operator, and exact V326 outer-domain α schedule. No secondary-backbone hyperparameter or α tuning was permitted. Each backbone produced 601,932 post-reference predictions over the same 586 cells and 12 complete domains. The independent verifier recomputed the domain effects and all deterministic certificates from the saved record-level predictions and passed 196 named checks.

Table S12. Zero-additional-tuning portability results. Differences are PCHP minus the corresponding protected-state MAE; negative values favor PCHP. Intervals are Bonferroni-adjusted two-sided 98.33% percentile intervals from complete-domain bootstrap resampling.

Backbone	Protected MAE	PCHP MAE	Difference	Adjusted interval	W/T/L
Ridge	0.11192	0.10773	−0.00419	[−0.00677, −0.00131]	10/0/2
HistGradientBoosting	0.07674	0.07249	−0.00425	[−0.00633, −0.00203]	11/0/1
LightGBM	0.07674	0.07252	−0.00422	[−0.00621, −0.00231]	11/0/1

All three pre-frozen conjunction gates passed. Maximum displacement and observed absolute-loss increase remained within the 0.01 budget, maximum protected-state and output increases were zero, outputs remained in $[0, 1.3]$, and exact prefix replay had maximum difference zero. Because all development outcomes had already been opened, this result excludes an ExtraTrees-specific explanation within the three audited families but does not establish external confirmation, universal model independence, deep-model portability, or best-backbone superiority.

Table S13. SHA-256 identifiers for the V375 portability audit.

Artifact	SHA-256
V375 pre-freeze JSON	ed6c7b6d119d4a3bb9111e27cae198c73f9c8590b3bb09756548ef06a95993e9
V375 audit program	d1924aa47c516e4f3474942e87044df7d5bec746531b8f3816244f610e92d5eb
V375 result report	862482438df32b59a19cfe8c4d15f413a11ee0baa2fd8e4451c9066e4d76562d
V375 independent verification receipt	26e81d0a73374a954e518dcda7e37bdd61ba04e88b407f8de55c82476a889376

9 Battery-sequence Transformer stress test

A compact pre-normalized Transformer (73,377 parameters; two encoder layers; four attention heads; model dimension 64) received voltage, C-rate, and normalized charge increment at 10s intervals over the first 600s. Three fixed source-only ensemble members were trained for 15 epochs on eight sequence-available development domains (206 cells; 20,425 balanced training rows) and evaluated without target-domain fine-tuning on complete SDU and MATR trajectories (255 cells;

187,864 records). The sequence proposals were projected using the authoritative V327 protected states, outer-domain α values, and $\delta = 0.01$. Target labels entered only the final scoring step.

Table S14. Zero-shot battery-sequence Transformer audit. The interval is a physical-cell bootstrap interval for PCHP minus the protected state.

Dataset	Transformer	Transformer+PCHP	Protected	Primary PCHP	PCHP-protected 95% interval
MATR	0.05311	0.05208	0.05079	0.04947	[0.00039, 0.00219]
SDU	0.14671	0.07381	0.06793	0.06529	[0.00484, 0.00688]

The Transformer candidate did not transfer as well as the protected tabular state. PCHP nevertheless reduced the larger proposal errors while enforcing the declared operating contract: maximum displacement and maximum realized absolute-loss increase were 0.01 within floating-point tolerance, and the audit recorded no budget, range, or monotonicity violation. This is evidence of architecture compatibility and harm control, not a claim that the tested Transformer improves prediction utility.

Table S15. SHA-256 identifiers for the V383 battery-sequence Transformer audit.

Artifact	SHA-256
Audit program	86476e41f23f6769d052db052a8d0ee1e5a56f80a31d7dab3bb0ecf23fb36b68
Result report	1acf00ad2191fe5b361f56c59ac4614b3138daca0d7e2706ddc7f5da4aafb247

10 Retrospective monotonicity-contract ablation

The V385 audit reused the 601,932 frozen development predictions from 12 held-out domains and 586 physical cells. The pointwise-tube arm removed only the previous-output cap: it projected the unchanged candidate independently into the same $[b_{it} - 0.01, b_{it} + 0.01]$ tube around the current strict protected state. It therefore isolates output monotonicity while holding the protected state, candidate, range, and harm budget fixed. The strict arm exactly reproduced the authoritative V327 predictions with maximum error zero. The bounded-recovery arm reused the V374 source-only-selected alternative contract, which allows both protected state and output to rise by at most 0.005 per emitted record; it is consequently an alternative state-and-output contract rather than a single-component deletion.

Table S16. Monotonicity-contract ablation. MAE is first averaged within physical cells and then equally across complete held-out domains. A positive transition is an output increase greater than 10^{-12} between consecutive records from one cell.

Contract	Domain MAE	Worst MAE	Rising steps	Cells	Max rise
Pointwise harm tube	0.070118	0.145307	38,580	483/586	0.020
Strict non-increasing PCHP	0.070378	0.145307	0	0/586	0
Bounded recovery	0.069584	0.149738	57,332	574/586	0.005

Relative to the pointwise tube, strict monotonicity changed domain-equal MAE by +0.000260; the paired complete-domain bootstrap 95% interval was $[-0.000538, 0.001095]$, the exact 2^{12} sign-flip value was 0.5654, and domain wins/ties/losses for the strict arm were 3/1/8. Thus the strict trajectory contract eliminated 38,580 upward revisions affecting 82.4% of cells without a clearly separated mean accuracy penalty. Bounded recovery changed MAE by -0.000794 relative to strict PCHP, but its interval $[-0.002708, 0.001041]$ and exact value 0.4492 did not separate the arms; its worst-domain MAE was 0.00443 higher. All three arms remained inside $[0, 1.3]$ and satisfied their arm-specific 0.01 displacement and observed absolute-loss certificates. The bounded-recovery maximum envelope excess was 4.34×10^{-18} . We therefore retain strict PCHP for the

stated immutable, non-increasing record contract; this retrospective ablation does not select a new deployment contract.

Table S17. SHA-256 identifiers for the V385 monotonicity-contract ablation.

Artifact	SHA-256
Frozen protocol	5ca2c74dc88fe6bdf18a797696eafdefad99b1c5156833b6a5a5510347c58b69
Audit program	802e29ad84f599f958ac690f009b89d0aff22eb4ff6382b0935a8afdc89828d3
Verification tests	d307adb08ce5165f72afa900942ca7c57bf06e3d6df3d58d8e7c7dd58254cc57
Result report	ef6026f091166183cde16cdacb87e746b76c66e5f3267187aec9123bc8af4ab5

11 Retrospective numerical-output-bound sensitivity

The V386 audit asked whether the declared numerical output range, rather than the learned candidate and trajectory contract, determined the reported results. It replayed the unchanged V321 projection operator on saved pre-projection predictions with upper bounds 1.1, 1.2, 1.3, and 1.5, while retaining the lower bound 0, source-selected assimilation rates, $\delta = 0.01$, cell order, and every scoring unit. No model was refitted and no bound was selected from accuracy. Development, the six external datasets, and BaSyTec were treated as target-compatible evidence surfaces. NASA was retained as a target-scale incompatibility stress test and was prohibited from selecting the bound.

For NASA, projection was replayed on all 2,598 label-blind prediction records from 34 cells before joining the 2,556 released labels from 33 scoreable cells. This preserves the original causal trajectory even where a record lacks a matched discharge label. At the reference upper bound 1.3, all four replays reproduced their saved protected states and PCHP outputs with maximum error zero.

Table S18. Numerical-output-bound sensitivity. The displayed MAE is domain-equal cell-macro MAE for development and the six external datasets, and cell-macro MAE for each single boundary surface. “Changed” counts outputs that differed from the 1.3 replay at any tested upper bound.

Surface	Truth max	Candidate max	Output max	MAE	Changed	Truth > 1.3
Development	1.059406	1.021457	1.014274	0.070378	0	0
Six external datasets	0.998467	0.981420	0.985900	0.126365	0	0
BaSyTec	1.004612	0.974823	0.907277	0.165101	0	0
NASA stress test	21.770609	0.932633	0.937209	0.727908	0	144/2,556

Every saved raw baseline and raw candidate on all four surfaces was below 1.1. Consequently, every replay produced exactly the same protected state, PCHP output, cell/domain MAE, and worst-domain MAE over the full 1.1–1.5 grid. All range, non-increasing-trajectory, 0.01 displacement, and observed absolute-loss checks passed. The compatible evidence therefore classifies the upper bound as **NONBINDING**. The retained value 1.3 is the pre-existing conservative computational bound, not a value tuned to improve the reported estimates. NASA’s 95th-percentile target of 1.4401 and maximum of 21.77 instead expose an incompatible normalization scale; enlarging an output bound on the basis of those labels would change the task definition rather than repair the model.

Table S19. SHA-256 identifiers for the V386 numerical-output-bound audit.

Artifact	SHA-256
Frozen protocol	aef0f74f7a4b2ac683214c0617021a02c4c0d8dd9a51e99856a716f6b080d0ba
Audit program	1e33f1249033a05564498a0e909982fd06f9ac90674b6f34193ba0e139ac73a2
Verification tests	c31f55285b35bc0be942caba2b4416216f5e6a8348f283d6942ff6146810a228
Result report	a44ae0e520f20197726cc1bf714a4035fe1b78a0204d8d92672d4e0e9588e424

12 Protocol-frozen NASA target-compatibility stress test

NASA PCoE Battery Data Set 5 was selected from official metadata and frozen before archive download [5]. The protocol fixed the official archive identity, recursive inclusion of MATLAB files, charge–discharge pairing, the 600s charge window, first-five reference, feature fields, $\alpha = 0.01$, $\delta = 0.01$, minimum cell size, estimand, decision gates, and hashes of the prediction and scoring programs. After duplicate identities were reconciled, 38 files represented 34 physical batteries.

The label-blind stage accessed cycle type and time plus voltage, current, temperature, and time arrays from charge records. It created and hashed 2,598 post-anchor predictions for all 34 batteries without reading discharge capacity. The independent scoring stage then read the frozen discharge locators and scalar **Capacity** field, formed the first-five-normalized target, and scored the prediction artifact once. The prespecified label audit rejected 44 non-scalar, non-finite, or non-positive discharge-capacity records. One predicted battery consequently lacked a valid first-five target reference, leaving 33 batteries and 2,556 scored records. No roster, reference, model, parameter, or endpoint changed after outcome access.

Across the 33 eligible batteries, battery-equal MAE was 0.73379 for the protected causal state and 0.72791 for PCHP. The paired difference was -0.00588 with a paired-battery bootstrap 95% interval of $[-0.00784, -0.00373]$; 27 batteries improved and 6 worsened. Maximum battery-level harm was 0.00849. Every point satisfied $|p_{it} - b_{it}| \leq 0.01$, the maximum observed absolute-loss increase was 0.01 within floating-point tolerance, and all protected and released trajectories were non-increasing.

The absolute target was not compatible with a high-accuracy capacity-retention claim. Battery B0041 had a first-five reference capacity of 0.05584 Ah and a maximum normalized value of 21.77; B0039 had a reference of 0.47114 Ah and a maximum normalized value of 3.76. Across all scored labels, the 95th percentile was 1.4401 and the maximum was 21.77, whereas the maximum saved candidate was 0.9326. The experiment therefore supports only a relative within-record harm-control result under severe target shift. It is not evidence of high-accuracy absolute SOH estimation, an independent-laboratory replication, or prospectively collected validation.

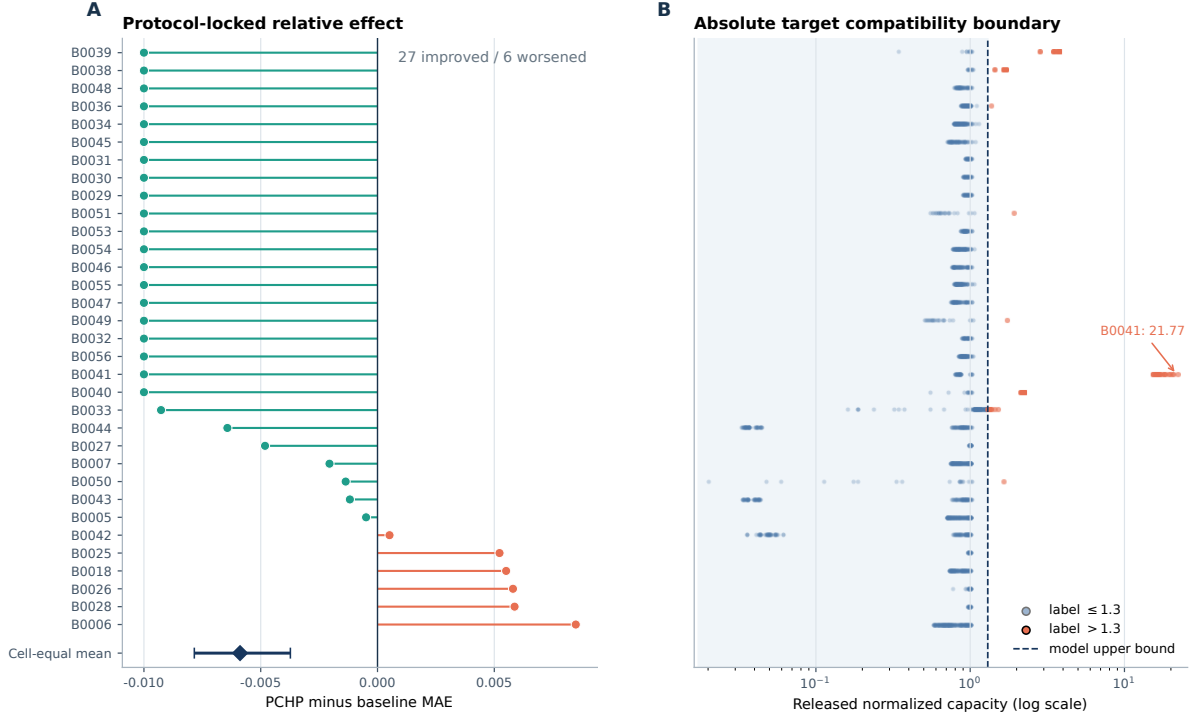


Figure S1. Protocol-frozen NASA target-compatibility stress test. **A**, paired PCHP-minus-protected-state MAE for all 33 eligible physical batteries; negative values favor PCHP, and the diamond with interval denotes the battery-equal mean and 95% paired-battery bootstrap interval from 100,000 resamples. **B**, released normalized-capacity labels by battery on a logarithmic axis, with the frozen model range $[0, 1.3]$ indicated. The 2,556 records are repeated measurements nested within batteries.

13 Protocol-frozen external mechanism and decision-cost confirmation

The V380 protocol froze six external battery datasets, 659 cells, 9,712 post-reference records, the complete-development ExtraTrees fit, $\alpha = 0.01$, $\delta = 0.01$, the exact development-selected $+0.01$ constant budget-feasible offset, the dataset-equal estimand, a two-stage dataset-cell bootstrap with 100,000 replicates and seed 20,260,803, deterministic certificate criteria, and the decision-cost grid. The archives had historical exposure in separate exploratory branches, but no external outcome or summary influenced PCHP, the comparator, estimand, cost protocol, or decision criteria. The appropriate classification is therefore protocol-frozen external mechanism confirmation on datasets that did not influence the tested design, not prospectively blinded data collection.

PCHP reduced dataset-equal MAE from 0.13326 for the constant offset to 0.12636. The full-precision primary result was

$$\Delta_{\text{ext}} = -0.0068971692,$$

$$\text{two-stage 95\% interval} = [-0.0119216689, -0.0021174927].$$

Five of six datasets improved, and the protected-state MAE was 0.1306078750. Maximum displacement and observed regret were 0.01 within floating-point tolerance; protected and projected trajectories had zero maximum increase; and independent prefix replay had zero maximum difference. The verifier passed all 49 named checks and classified mechanism inference and deterministic certification as PASS.

A post-hoc robustness audit then reused the unchanged V380 prediction ledger without fitting, tuning, threshold selection, or dataset removal. The complete external dataset remained the top-level unit; cells and records were treated as nested observations. Table S20 separates changes in the

target population induced by weighting from evidence about the frozen dataset-equal estimand. Negative values favor PCHP.

Table S20. External aggregation sensitivity. Each entry is PCHP MAE minus comparator MAE. The exact p value enumerates all $2^6 = 64$ sign assignments of the six dataset-level cell-macro effects. Only the constant-offset comparison is the primary exact test; the protected-state result is descriptive.

Comparator	Dataset mean	Cell mean	Record mean	Dataset median	Cell median	Exact p
Constant offset	−0.006897	−0.007367	−0.008864	−0.007143	−0.004226	0.0625
Protected state	−0.004243	−0.001943	−0.002507	−0.005912	−0.003373	0.1250

The direction did not depend on one external dataset. For PCHP versus the constant offset, all six leave-one-dataset-out means were negative, ranging from −0.008653 to −0.004798; after removing ISU–ILCC, the largest dataset by both cell and record count, the mean remained −0.006567. The corresponding protected-state range was [−0.005517, −0.003411], with a largest-dataset-removed effect of −0.005207. Thus the external advantage is stable to the declared aggregation and omission checks. With only six top-level units and one dataset-level loss, however, the two-sided exact sign-flip value for the primary comparison is 0.0625; this sensitivity is reported as effect-direction evidence rather than a separate 0.05-level confirmation claim.

Table S21. Dataset-equal continuous asymmetric decision cost. The missed-degradation term receives weight r relative to unnecessary review.

r	Protected state	Constant offset	PCHP
1	0.13061	0.13326	0.12636
2	0.21527	0.22425	0.20970
5	0.46925	0.49721	0.45970
10	0.89254	0.95214	0.87636

At the primary $\tau = 0.80$ and $r = 5$, binary review cost was identical across methods because no PCHP update crossed the action threshold. The verifier therefore retained the prespecified PASS_NON_CLAIM status for binary decision benefit. The supported decision result is the 7.54% continuous-cost reduction from the constant offset to PCHP.

The observed-horizon event audit defined late warning as a predicted first crossing after the observed crossing or no predicted crossing within follow-up. Premature review was a predicted crossing before the observed crossing; for cells without an observed crossing, any predicted crossing within follow-up was premature. An absent predicted crossing was censored to one record beyond the final observation for restricted-delay calculation. Table S22 reports the primary 80% capacity-retention threshold. Metrics are dataset-equal; rates are calculated within physical cells before aggregation.

Table S22. Observed-horizon first-crossing metrics at $\tau = 0.80$. All three methods issued identical threshold actions.

Method	Restricted absolute delay	95th-percentile late delay	Absolute cycle delay	Late-warning cell rate	Premature-review rate ^c
Protected state	9.32	16.49	1206.33	0.914	0.086
Constant offset	9.32	16.49	1206.33	0.914	0.086
PCHP	9.32	16.49	1206.33	0.914	0.086

^cAmong cells with an observed true crossing. Delays are in record-index units except “absolute cycle delay,” which uses the source cycle-number coordinate.

Sensitivity thresholds led to the same operational conclusion. At $\tau = 0.70$, PCHP and the constant offset had identical first-crossing delays and cell-level late/premature rates; PCHP increased

the dataset-equal premature-review record rate by 0.00489. At $\tau = 0.90$, PCHP reduced restricted absolute delay by only 0.00357 records and the cell-macro late-warning record rate by 0.00071, without changing the dataset-equal late-warning or premature-crossing cell rates. The current budget therefore changes continuous error severity but not threshold timing at a practically material scale.

Table S23. SHA-256 identifiers for the V383 threshold-event audit.

Artifact	SHA-256
Audit program	92d28a2e5f5a8c3826d95a99165e616c285fc5872d45d7d0c9bc521bd00c3264
Result report	19122af913409a4c2f0254e7604ccb288926253a22653e1ef61fb18336275830
Cell-metric table	591569b17bcfad5632d9acc370bf592bbc3e8c353df56cc8998e1a692980bd9c
Dataset-equal summary	df427571293dbcf67191dc07f1db132f00837f65b989db24b86e0519df15089f

Table S24. SHA-256 identifiers for the V380 mechanism confirmation and V384 external robustness audit.

Artifact	SHA-256
V380 pre-freeze protocol	128fa29a2b21ad8022adf3afecc6351e8b480e770c4c8833a032c7097575b5cc
V380 execution program	9a4e819a20f1053dc2ae447a396800fa4b88af49a101de1692503f18d4d0deb1
V380 independent verifier	479533c75f362bd64f9d00c44582612ab39e994ba08d16d04bdb886990831830
Frozen predictions	474d01b7473ed30aaa79f117ab1a9237e5028201ecab74dbce37932d3ac9f5e3
Dataset metrics	482b6f3717f656c6fcd74ee932aae172c8a5029feebdff5ba21757478ded27b2
Decision-cost summary	4fa9c0716b2fc9c4b56e0c03699207e3cad16aa7fff1e57747d603cf3e2c2d74
V380 result report	e46f1157791f8da33e401833327dc2dba791a43df496a2d7204a932cf25c4655
Independent verification receipt	798406ec12cbd8ce0c89a12779cdb7b389bd06f52f76d04cb6876d08964b45c3
V384 robustness protocol	0d10e30f6497be482b074a89cd3bb3e8c8d6f3b2864fa9453ea9bf7e0b73c517
V384 audit program	e80ed65a64b68d137da436aebcaa606d2e7263270843d3c98ab6dfd931e0c54d
V384 verification tests	d0611d66c8281c5d9943579d0f0603c4feafab14c4ad1545ce9d7ffa2810084e
V384 result report	60a8dc4d1dfa440c2815cfc05ae523c113dfc3f3cf759b1c28e698ee0c5a5832

14 Analytical proofs and loss-contract extensions

14.1 Protected-state and projection proofs

The protected-state update is prefix invariant because b_{i1} uses only r_{i1} , and every subsequent b_{it} is a deterministic function of $(b_{i,t-1}, r_{it})$. Induction therefore proves that extending a trajectory after record t cannot alter its first t states. For $t > 1$, the same update has the exact proximal form

$$b_{it} = \arg \min_{q \in \mathcal{Y}, q \leq b_{i,t-1}} \left\{ \frac{\alpha}{2}(q - r_{it})^2 + \frac{1 - \alpha}{2}(q - b_{i,t-1})^2 \right\}. \quad (1)$$

The unconstrained minimizer is $(1 - \alpha)b_{i,t-1} + \alpha r_{it}$; projection onto the one-sided feasible set replaces r_{it} by $\min\{r_{it}, b_{i,t-1}\}$, followed by clipping to $\mathcal{Y}$, which is exactly the protected recursion.

Recursive feasibility and trajectory contract. At the first post-reference record, $p_{i0} = 1.3$ and $b_{i1} \in [0, 1.3]$, so the lower endpoint $\max\{0, b_{i1} - \delta\}$ cannot exceed any term defining the upper endpoint. Suppose the interval was feasible at $t - 1$. Since $b_{it} \leq b_{i,t-1}$ and $p_{i,t-1} \geq \max\{0, b_{i,t-1} - \delta\}$,

$$p_{i,t-1} \geq \max\{0, b_{it} - \delta\}.$$

The current interval is therefore nonempty. Projection gives range membership and $|p_{it} - b_{it}| \leq \delta$; the previous-output cap gives $p_{it} \leq p_{i,t-1}$; and prefix measurability of every input gives prefix causality.

Prefix nonexpansiveness. Write

$$F_\alpha(x, z) = \Pi_{\mathcal{Y}}((1 - \alpha)x + \alpha \min\{x, z\}).$$

Minimum is jointly one-Lipschitz in the sup norm, and convex combination plus scalar clipping preserves this constant. Initialization and induction yield

$$\max_{s \leq t} |b_{is} - \tilde{b}_{is}| \leq \max_{s \leq t} |r_{is} - \tilde{r}_{is}|.$$

Define $L(b) = \max\{0, b - \delta\}$ and $U(b, u) = \min\{1.3, b + \delta, u\}$. The released value is

$$p_{it} = \text{med}\{L(b_{it}), c_{it}, U(b_{it}, p_{i,t-1})\}.$$

Minimum, maximum, and median are jointly one-Lipschitz in the sup norm. Thus

$$|p_{it} - \tilde{p}_{it}| \leq \max\{|b_{it} - \tilde{b}_{it}|, |c_{it} - \tilde{c}_{it}|, |p_{i,t-1} - \tilde{p}_{i,t-1}|\},$$

and induction proves the prefix-level bound in the main text. Two interior candidates separated by $\varepsilon > 0$, with all other inputs equal, produce outputs separated by exactly ε ; the constant one is therefore tight.

Exact absolute-loss region, viability, and closest-candidate optimality. The triangle inequality gives

$$|p - y| - |b - y| \leq |p - b|$$

for every y , and equality is attained at $y = b$. Consequently the outcome-uniform absolute-loss region is exactly $[b - \delta, b + \delta]$, and intersecting it with the numerical range and previous-output half-line gives the main-text interval I_{it} . At $\delta = 0$, any $p \neq b$ incurs additional loss $|p - b| > 0$ at $y = b$, proving the zero-budget limit.

For any $q \in I_{it}$ and any future non-increasing protected-state path, let $L_{is} = \max\{0, b_{is} - \delta\}$. This lower-endpoint path is non-increasing, remains within the range and harm budget, and begins no higher than q ; it is therefore a feasible continuation. A point outside I_{it} already violates a current constraint and cannot be extendable. Finally, over the exact feasible set I_{it} , the strictly convex objective $\frac{1}{2}(p - c_{it})^2$ has the unique minimizer $\Pi_{I_{it}}(c_{it})$.

14.2 Extensions to other loss contracts

Metric loss. For any metric space $(\mathcal{M}, d)$,

$$\sup_{y \in \mathcal{M}} [d(p, y) - d(b, y)] = d(p, b). \quad (2)$$

The upper bound is the triangle inequality and equality follows at $y = b$. Hence the exact harm region is the closed metric ball of radius δ around b . This is an established metric consequence; its role here is to expose the geometry underlying the deployment contract.

Squared loss and the declared outcome set. For $L < U$, $b, p \in [L, U]$, and $\ell_2(q, y) = (q - y)^2$,

$$\sup_{y \in [L, U]} [(p - y)^2 - (b - y)^2] = \begin{cases} (p - b)(p + b - 2L), & p \geq b, \\ (b - p)(2U - p - b), & p < b. \end{cases} \quad (3)$$

The difference expands to $(p - b)(p + b - 2y)$, an affine function of y ; the maximizing endpoint therefore switches with the sign of $p - b$. Solving the two budget inequalities gives the exact region

$$\mathcal{H}_\delta^{(2)}(b; [L, U]) = \left[\max\left\{L, U - \sqrt{(U - b)^2 + \delta}\right\}, \min\left\{U, L + \sqrt{(b - L)^2 + \delta}\right\} \right]. \quad (4)$$

If the outcome set is instead $\mathbb{R}$, every $p \neq b$ has infinite worst-case additional squared loss: for $p > b$ the difference diverges as $y \rightarrow -\infty$, and for $p < b$ it diverges as $y \rightarrow +\infty$.

Let the two endpoints in Eq. (4) be $\underline{q}_\delta^{(2)}(b)$ and $\bar{q}_\delta^{(2)}(b)$. The lower endpoint is nondecreasing in b . Therefore, for a non-increasing protected state,

$$I_{it}^{(2)} = [\underline{q}_\delta^{(2)}(b_{it}), \min\{\bar{q}_\delta^{(2)}(b_{it}), p_{i,t-1}\}]$$

is nonempty by the same induction as the absolute-loss interval and certifies squared-loss increase no greater than δ for every $y \in [L, U]$. This extension is valid only when $[L, U]$ is a credible outcome contract.

Directionally weighted absolute loss. For positive costs c_{under} and c_{over} , define

$$\ell_{\text{asym}}(p, y) = c_{\text{under}}(y - p)_+ + c_{\text{over}}(p - y)_+.$$

Direct case analysis gives

$$\sup_{y \in \mathbb{R}} [\ell_{\text{asym}}(p, y) - \ell_{\text{asym}}(b, y)] = \begin{cases} c_{\text{over}}(p - b), & p \geq b, \\ c_{\text{under}}(b - p), & p < b. \end{cases} \quad (5)$$

Thus a decision-loss budget η yields the exact directional interval

$$\left[b - \frac{\eta}{c_{\text{under}}}, b + \frac{\eta}{c_{\text{over}}} \right].$$

Intersecting this interval with $[0, 1.3]$ and the previous-output cap preserves recursive feasibility when costs and budget are constant along a trajectory. Equal costs and $\eta = \delta$ recover the experimental PCHP contract.

If costs and budget vary with t , define

$$L_{it} = \max \left\{ 0, b_{it} - \frac{\eta_{it}}{c_{\text{under},it}} \right\}, \quad U_{it} = \min \left\{ 1.3, b_{it} + \frac{\eta_{it}}{c_{\text{over},it}} \right\}.$$

The realized interval $[L_{it}, \min\{U_{it}, p_{i,t-1}\}]$ is nonempty exactly when $L_{it} \leq p_{i,t-1}$. It is feasible for every candidate sequence exactly when $L_{it} \leq L_{i,t-1}$ for all $t \geq 2$. Given the already issued output, the least viable current budget is

$$\eta_{it}^{\min} = c_{\text{under},it}(b_{it} - p_{i,t-1})_+. \quad (6)$$

A smaller declared budget must trigger an explicit failure rather than silent budget enlargement.

Aggregation. For nonnegative weights w_k , record-level guarantees imply

$$\sum_k w_k (|p_k - y_k| - |b_k - y_k|) \leq \sum_k w_k \delta_k.$$

No independence assumption is required for this deterministic inheritance. Statistical uncertainty in empirical utility remains governed by the domain–cell–record sampling hierarchy.

15 Loss-geometry verification record

The V366 extension was frozen before numerical execution. Its deterministic validator read no battery outcome and used seed 20,260,802 with tolerance 10^{-12} . It verified the metric identity, the exact bounded squared-loss supremum and feasible interval, active-boundary equality, adjacent exterior-point violation, recursive feasibility, and the full-real-line squared-loss divergence construction.

All 27 aggregate checks passed. Across 200,000 random bounded configurations, the maximum formula discrepancy was 1.3323×10^{-15} . Across 2,000 non-increasing trajectories of length 128, the maximum budget excess was 7.7716×10^{-16} and the maximum output increase was zero. These tests target transcription and implementation errors; the analytical proofs remain the scientific basis of the claims.

Table S25. SHA-256 identifiers for the pre-frozen V366 loss-geometry extension.

Artifact	SHA-256
V366 loss-geometry pre-freeze	3e9b0494c23d10500889a27811fe8b9c29de74e60712271467e4d0c28ad0d1d0
V366 deterministic validator	6961dcf3d141397b07e6890d952d52e60689209a7386395d3f51e54e2557a344
V366 validation report	72cbb80861c9fc456a2aa7ba857757a33f3eb3585b7b9121366f70894476c8b0
V366 claim-source audit	4c430aca07747dc4651173c4458a69fc664d4fae4d684071fe0f97fe5dc2308a

16 Prefix nonexpansiveness verification record

The V377 operator-stability protocol was frozen before numerical execution. The validator read no battery outcomes and used seed 20,260,802 with absolute tolerance 2×10^{-12} . It completed 3,888 exhaustive local protected-state checks, 21,168 exhaustive local projection checks, 108 proximal-equivalence checks, and 30,000 independently generated trajectory-pair checks. The maximum protected-state inequality excess was 1.1102×10^{-16} ; the maximum complete-output inequality excess was 3.2818×10^{-16} ; and the explicit witness attained the input separation 0.03 exactly up to floating-point representation. The report recorded a pass. These checks target implementation and proof-transcription failures; the theorem remains supported by the analytical proof above.

Table S26. SHA-256 identifiers for the V377 operator-stability verification.

Artifact	SHA-256
V377 deterministic validator	9abe330253305b02250bba00da337ec8003fe5c9aa99b535a4af2cb842e38b35
V377 validation report	2421dd474feb23a349b3be01354de240baf7b945ce5bbcdc95d5c7edbbd683f8
V377 four-test extension	66f2fc349826e6e9c1fa22529a99546e5a30405dbbc7ee11b4dc8f2c1c76f14e

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